update_dns_ipv6.md

Updating DNS Configuration After IPv6 Prefix Changes

This document provides a clear, technically complete workflow for updating BIND9 DNS configuration when your IPv6 delegated prefix changes. It expands on the basic hints and includes best practices for zone hygiene, serial management, and safe reload/notify sequencing across primary and secondary servers.

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1. Update ACLs and Options for the New Delegated Prefix

Modify your primary server's named.conf.options (or equivalent) so that any ACLs referencing your IPv6 prefix include the newly delegated range.
Example:

```
acl "internal-v6" {
2001:db8:abcd:1234::/56; // update to new prefix
};
```

After editing, validate syntax:

```
named-checkconf
```

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2. Update home.lan Forward Zone

Update all AAAA records that depend on the old prefix. Also increment the SOA serial.
Example SOA bump:

```
@ IN SOA ns1.home.lan. admin.home.lan. (
2026051701 ; serial - increment
3600 ; refresh
900 ; retry
604800 ; expire
86400 ) ; minimum
```

Validate the zone:

```
named-checkzone home.lan /etc/bind/zones/home.lan
```

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3. Update ipv6.home.lan.rev Reverse Zone

Update PTR entries to reflect the new IPv6 prefix and increment the SOA serial.
Validate the reverse zone:

```
named-checkzone ipv6.home.lan.rev /etc/bind/zones/ipv6.home.lan.rev
```

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4. Update named.conf.local on Primary

Ensure the also-notify IPv6 address points to the correct secondary server.

Example:

```
also-notify { 2001:db8:abcd:1234::2; }; // update to new secondary IPv6
```

Validate configuration:

```
named-checkconf
```

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5. Reload Primary DNS

Apply all changes on the primary:

```
sudo rndc reload
```

Check logs for errors:

```
sudo journalctl -u bind9 -n 50
```

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6. Update Secondary Server

On the secondary:

- Update named.conf.local so the masters { ... } block references the primary's new IPv6 address.
- Reload configuration:

```
sudo rndc reload
```

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7. Trigger Notify from Primary

Force the primary to notify the secondary of zone changes:

```
sudo rndc notify home.lan
```

Confirm notify events in logs.

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8. Refresh Secondary

On the secondary, explicitly request a refresh:

```
sudo rndc refresh
```

Check that the new serial is pulled:

```
sudo rndc zonestatus home.lan
```

Recommended Additional Safety Checks

- Confirm both servers now serve the same SOA serial.
- Use dig to verify forward and reverse resolution:

```
dig AAAA host.home.lan @primary
dig -x 2001:db8:abcd:1234::50 @secondary
```

- Ensure firewall rules allow DNS/IPv6 traffic between primary and secondary.

This checklist ensures a clean, consistent update cycle whenever your IPv6 delegated prefix changes, minimizing stale records and synchronization issues across your DNS infrastructure.